

Module 21 – KNN Case Study on Diabetes

Objective

In this module, you will apply the **K-Nearest Neighbors (KNN)** algorithm to build a **Diabetes Prediction Model**. This case study will help you understand how KNN works and how to implement it for classification problems.

Reference

Watch the **Module 21 video** carefully. The concepts and steps explained there will guide you in completing this case study.

Dataset

You can use the **Diabetes dataset** available on [Kaggle](#)

Task Instructions

1. Import and explore the dataset (check for missing values, data types, etc.).
2. Perform data preprocessing (handle missing data, normalize features if necessary).
3. Split the dataset into **training** and **testing** sets.
4. Implement the **K-Nearest Neighbors (KNN)** algorithm for prediction.
5. Evaluate the model using **accuracy**, **confusion matrix**, and other suitable metrics.
6. Provide a short analysis of your results.

Submission Requirements

- Submit your **.ipynb** file (Jupyter Notebook).
- Upload your work to **Google Colab** and **share the Colab link** in your submission.
- Ensure the shared link has “**Anyone with the link can view**” access.

Deliverables

- **KNN_Diabetes.ipynb** file
- Google Colab share link